

Daniel Sainati

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EDUCATION

- **University of Pennsylvania**
Doctor of Philosophy in Computer Science
Advisors: Benjamin C. Pierce and Stephanie Weirich
August 2024 - Ongoing
Philadelphia, Pennsylvania
- **Cornell University**
Master of Engineering in Computer Science
Advisor: Adrian Sampson
December 2017 - May 2018
Ithaca, New York
- **Cornell University**
Bachelor of Science in Computer Science
August 2014 - December 2017
Ithaca, New York

EMPLOYMENT

- **Senior Programming Language Engineer**
Flow Foundation
October 2021 - July 2024
Vancouver, British Columbia
 - Worked on **Cadence**, a high level statically-typed smart contract language
- **Software Engineer**
Meta
August 2018 - October 2021
Seattle, Washington
 - Worked on **Flow**, a static type system for JavaScript, used at Meta on tens of millions of lines of code
- **Software Engineering Intern**
Meta
Summer 2017
Menlo Park, California
 - Worked on **HHVM**, Meta's just-in-time Hack compiler
- **Software Engineering Intern**
Two Sigma Investments
Summer 2016
New York, New York

PUBLICATIONS

REFEREED CONFERENCE, JOURNAL, AND WORKSHOP PUBLICATIONS

- **Modular GPU Programming with Typed Perspectives**. Manya Bansal, **Daniel Sainati**, Joseph W. Cutler, Saman Amarasinghe, and Jonathan Ragan-Kelley. In [Programming Language Design and Implementation \(PLDI\) 2026](#) (**Distinguished Paper**). 25 pages.
- **The Search for Constrained Random Generators**. Harrison Goldstein, Hila Peleg, Cassia Torczon, **Daniel Sainati**, Leonidas Lampropoulos, and Benjamin C. Pierce. In [Programming Language Design and Implementation \(PLDI\) 2026](#). 26 pages.
- **Typing Strictness**. **Daniel Sainati**, Joseph W. Cutler, Benjamin C. Pierce, and Stephanie Weirich. In [Principles of Programming Languages \(POPL\) 2026](#). 27 pages.
- **LambdaLab: An Interactive λ -Calculus Reducer for Learning**. **Daniel Sainati** and Adrian Sampson. In [SPLASH-E 2018](#). 10 pages.

BOOKS

- **Software Foundations in Lean**. Mike Hicks, Benjamin C. Pierce, and the **Software Foundations in Lean Team**.
 - Among the primary translators of the original [Software Foundations](#) into Lean across three volumes: *Logical Foundations in Lean*, *Hoare Logic in Lean*, and *Type Systems in Lean*

HONORS, FUNDING, AND AWARDS

- **NSF CSGrad4US Fellowship** 2023
National Science Foundation
- **eBay Undergraduate TA Award** 2016
Cornell University
- **Engineering Dean's List** 2014-2017
Cornell University

TALKS

- **Modular GPU Programming with Typed Perspectives** September 2026
Brown Cognitive Engineering Lab (invited) Providence, Rhode Island
- **Modular GPU Programming with Typed Perspectives** September 2026
MathWorks Code Generation Knowledge Forum (invited) Natick, Massachusetts
- **Typing Strictness** February 2026
New York University Formal Methods Seminar (invited) New York, New York
- **Typing Strictness** February 2026
Portland State University PLV Seminar (invited) Portland, Oregon
- **Typing Strictness** December 2025
Cornell Programming Languages Discussion Group (invited) Ithaca, New York
- **Typing Strictness and Laziness with Effects and Coeffects** May 2025
New Jersey Programming Languages and Systems Seminar New York, New York

TEACHING AND SERVICE

- **Co-organizer** May 2026
New Jersey Programming Languages and Systems Seminar Philadelphia, Pennsylvania
- **Co-organizer** 2026
PLClub @ University of Pennsylvania Philadelphia, Pennsylvania
- **Student Volunteer**
 - January 2026 / POPL 2026 / Rennes, France
- **Teaching Assistant**
University of Pennsylvania
 - Fall 2026 / CIS 5000 / Software Foundations
 - Fall 2025 / CIS 5000 / Software Foundations
- **Teaching Assistant**
Cornell University
 - Spring 2018 / CS 6110 / Advanced Programming Languages
 - Fall 2017 / CS 4810 / Introduction to Theory of Computation
 - Spring 2017 / CS 3152 / Introduction to Computer Game Development
 - Fall 2015 - Fall 2016 / CS 3110 / Data Structures and Functional Programming (Fall 2016 Head TA)
 - Spring 2015 / CS 2110 / Object-Oriented Programming and Data Structures